SHAGITHTHIAH KIRUPAKARAN

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Research Interests

Machine Learning, Computer Vision, IoT Systems, AI for Disaster Response, Edge Computing, Data-Centric Machine Learning, Computer Systems, Heterogeneous Computing, Computer Architecture, Performance Engineering, Resource Management

Education

- **University of Peradeniya** *Expected Graduation: June 2028*
  Bachelor of the Science of Engineering(BScEng), GPA: 3.64/4.00 Kandy, Sri Lanka

Research Experience

- **Crisis Lab** *Apr 2026 - Present*
  Research Intern (Remote) Wellington, New Zealand
  - Curated and annotated ∼7,500 ground-level flood images from the NIWA Citizen Science API for flood-severity analysis, using an LLM-assisted workflow for damage relevance and severity labeling.
  - Fine-tuned ResNet18 for 4-class flood-severity classification, achieving ∼73% validation accuracy with checkpointing and learning-rate scheduling.
  - Contributing to the design of a citizen-science reporting architecture integrating ML-based severity classification, geo-tagged reports, and a responder dashboard.
- **Multidisciplinary AI Research Centre (MARC)** *May 2025 - Jul 2025*
  Research Volunteer
  - Contributed to a machine learning-based slum detection project, utilizing high-resolution satellite imagery to identify and map informal settlements.
  - Annotated and verified spatial data using Google Earth Pro, cross-referencing satellite and street-level views to construct a reliable ground-truth dataset for model training.

Final-Year Research Project

- **Cross-Resource Interference in Heterogeneous ARM Systems** *Sep 2026 - Present*
  Linux, Python, C/C++, Bash, perf, Hardware Performance Counters (PMU), NVIDIA Profiling Tools Research Project
  - Supervised by Dr. Asitha Bandaranayake, Head of the Department, Department of Computer Engineering, University of Peradeniya.
  - Investigating CPU–GPU–memory interference on heterogeneous ARM/Linux systems by co-running compute- and memory-intensive workloads and measuring execution time, IPC, cache misses, memory bandwidth, CPU stalls, and GPU utilization.
  - Designing reproducible interference experiments using `perf`, PMU counters, CPU affinity, workload co-execution, and GPU profiling to identify shared-resource bottlenecks and quantify performance degradation.
  - Working toward validation on Nanopore genomics pipelines spanning GPU-intensive basecalling and CPU/memory-intensive downstream processing.

Selected Projects

- **SmartIV - Intelligent IV Drip Monitoring System (Group Project)** *Jan 2026 - Present*
  React Native, AWS IoT Core, MQTT, Tauri (Rust) End-to-End System
  - Co-developed an IoT-based IV-drip monitoring system integrating bedside hardware, MQTT communication, AWS IoT Core, and a React Native application for real-time monitoring, alerts, and session history.
  - Owned the mobile/cloud integration, implementing the React Native interface, AWS IoT/MQTT data path, Cognito authentication, and backend deployment for device-to-application telemetry.
- **Voting Dynamics Using Agent-Based Modelling (Personal Project)** *Dec 2025*
  Python, NumPy, NetworkX, Matplotlib
  - Built an Agent-Based Model simulating how opinions spread and polarize across social networks.
  - Modeled voter behavior using social influence, noise, and zealotry as key parameters.
  - Found that even a small group of zealots can block consensus, mirroring real-world social-media polarization.
- **EngEx 2025 Registration Portal - Smart Exhibition Management System (Group Project)** *Aug 2025 - Nov 2025*
  React, Node.js, Express, PostgreSQL 2nd Year Project
  - Built an RFID-based visitor registration portal for EngEx 2025, the university's annual engineering exhibition.
  - Developed the Authentication & Registration module: secure login, JWT sessions, and individual/batch visitor registration.
  - Integrated the module with the RFID tag-linking system for real-time use during the live exhibition.

Technical Skills

ML/Data: PyTorch, NumPy, NetworkX, Matplotlib
Programming: Python, C/C++, Bash, Rust
Cloud/IoT: AWS IoT Core, MQTT, AWS Cognito, React Native
Systems: Linux, perf, PMU / hardware performance counters, NVIDIA profiling tools

Achievements & Awards

- **Winner - SLASSCOM National Ingenuity Awards 2026** *Jun 2026*
  Best Innovative Product (University Category) - Central Province
  - Awarded for Smart IV, an IoT-based system designed to monitor IV drip flow in real-time, estimate remaining volume, and instantly alert healthcare staff.
- **1st Runner-Up - NetX IoT Challenge** *Sep 2026*
  Team Zephyrus - IEEE Student Branch, University of Sri Jayewardenepura
  - Secured 1st Runner-Up at the NetX IoT Challenge with Smart IV Project.
- **Finalist - Innovo IoT Challenge 2026** *Sep 2026*
  SLIIT
  - Selected as a finalist with the Smart IV project.
- **Finalist - SLIoT Challenge** *Apr 2026*
  Smart IV Project
  - Presented the Smart IV prototype at the national finals.
- **Finalist - Startup Spark, TECHNO 2024** *Oct 2024*
  Intelligent Traffic Monitoring System (ITMS)
  - Pitched an AI-driven traffic management solution featuring dynamic signal control, emergency vehicle prioritization, and real-time accident detection to industry experts and potential investors.
- **University Colours - Chess** *2024*
  University of Peradeniya
  - Awarded University Colours in recognition of outstanding performance as a key member of the Women's Chess Team that secured the Championship title at the Inter-University Games 2024.

Extra Curricular Activities

- **National Arbiter** *2025 - Present*
  Chess
  - Passed the National Arbiter certification exam in 2022 and officially awarded the title in 2025 after successfully fulfilling all required tournament norms.
- **University Team Player** *Jun 2024 - Apr 2025*
  Women's Chess Team, University of Peradeniya
  - Secured 1st Place at the Inter-University Championship 2024, representing the university and collaborating on team strategies and match preparations.
- **Captain** *2024 - Present*
  E21 Batch Women's Chess Team
  - Led the batch team to a Runners-Up finish at the Freshers Meet 2024, organizing practice sessions and fostering a collaborative environment for intra-faculty championships.

Teaching Experience

- **Undergraduate Teaching Assistant** *Dec 2024 - Mar 2025*
  CO1010 Programming for Engineers Department of Computer Engineering, UoP
  - Supported undergraduate instruction in Python and object-oriented programming through hands-on guidance and problem-solving assistance.

References

- **Dr. Asitha Bandaranayake**
  Head of the Department & Senior Lecturer | FYP Supervisor Department of Computer Engineering, University of Peradeniya
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